Question	Answer	Marks
1(a)(i)	radius = $6.37 \times 10^6 \times \cos 52.2° = 3.90 \times 10^6$ m	**A1**
1(a)(ii)	period = 24 hours	**C1**
	$v = 2\pi r / T$ **or** $v = r\omega$ **and** $\omega = 2\pi / T$	**C1**
	$v = (2\pi \times 3.90 \times 10^6) / (24 \times 60 \times 60)$ $= 280$ m s^{-1}	**A1**
1(b)(i)	$F = mv^2 / r$	**C1**
	$= (58.6 \times 280^2) / (3.90 \times 10^6)$ $= 1.2$ N	**A1**
1(b)(ii)	arrow pointing horizontally to the left	**B1**
1(b)(iii)	arrow from student pointing along the dotted line, labelled ‘weight’	**B1**
	upwards arrow from student pointing in a direction to the left of normal and above the tangent to the Earth, labelled ‘contact force’	**B1**

Question	Answer	Marks